Paths in a dependent pair type: the path type is equivalent to the dependent sum of a base path and a path over it

The Univalent Foundations Program

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Editorial note. The counted statement is the author theorem labelled transport-Sigma sibling thm:path-sigma (source lines 1426-1431); the proof span 1433-1480 is the author's proof environment, retained line for line.

Editorial note. The statement and proof cite no external labels, so no external references are declared.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the characterization of paths in dependent pair types. Only the counted theorem and its author proof are retained; the surrounding remarks and the uniqueness-principle corollary remain uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent pair types (Sigma types) with their induction principle (pair induction).
- Transport along a path in a type family, with its definitional behaviour on reflexivity.
- Dependent paths over a base path: paths in the total space project to base paths with a lifting condition.
- The induction principle for identity types (path induction) and equivalences via quasi-inverse.

Theorem 0.0.1. *Suppose that $P : A \to \mathcal{U}$ is a type family over a type A and let $w, w' : \sum_{(x:A)} P(x)$. Then there is an equivalence*

$$(w = w') \simeq \sum_{(p:\mathsf{pr}_1(w)=\mathsf{pr}_1(w'))} p_*(\mathsf{pr}_2(w)) = \mathsf{pr}_2(w').$$

Proof. We define a function

$$f : \prod_{w,w':\sum_{(x:A)} P(x)} (w = w') \to \sum_{(p:\mathsf{pr}_1(w)=\mathsf{pr}_1(w'))} p_*(\mathsf{pr}_2(w)) = \mathsf{pr}_2(w')$$

by path induction, with

$$f(w, w, \mathsf{refl}_w) :\equiv (\mathsf{refl}_{\mathsf{pr}_1(w)}, \mathsf{refl}_{\mathsf{pr}_2(w)}).$$

We want to show that f is an equivalence.

In the reverse direction, we define

$$g : \prod_{w,w' : \sum_{(x:A)} P(x)} \Big(\sum_{p : \mathsf{pr}_1(w) = \mathsf{pr}_1(w')} p_*(\mathsf{pr}_2(w)) = \mathsf{pr}_2(w') \Big) \to (w = w')$$

by first inducting on w and w', which splits them into (w_1, w_2) and (w_1', w_2') respectively, so it suffices to show

$$\Big(\sum_{p : w_1 = w_1'} p_*(w_2) = w_2' \Big) \to ((w_1, w_2) = (w_1', w_2')).$$

Next, given a pair $\sum_{(p:w_1 = w_1')} p_*(w_2) = w_2'$, we can use Σ-induction to get $p : w_1 = w_1'$ and $q : p_*(w_2) = w_2'$. Inducting on p, we have $q : (\mathsf{refl}_{w_1})_*(w_2) = w_2'$, and it suffices to show $(w_1, w_2) = (w_1, w_2')$. But $(\mathsf{refl}_{w_1})_*(w_2) \equiv w_2$, so inducting on q reduces the goal to $(w_1, w_2) = (w_1, w_2)$, which we can prove with $\mathsf{refl}_{(w_1, w_2)}$.

Next we show that $f(g(r)) = r$ for all w, w' and r, where r has type

$$\sum_{(p : \mathsf{pr}_1(w) = \mathsf{pr}_1(w'))} (p_*(\mathsf{pr}_2(w)) = \mathsf{pr}_2(w')).$$

First, we break apart the pairs w, w', and r by pair induction, as in the definition of g, and then use two path inductions to reduce both components of r to refl. Then it suffices to show that $f(g(\mathsf{refl}_{w_1}, \mathsf{refl}_{w_2})) = (\mathsf{refl}_{w_1}, \mathsf{refl}_{w_2})$, which is true by definition.

Similarly, to show that $g(f(p)) = p$ for all w, w', and $p : w = w'$, we can do path induction on p, and then pair induction to split w, at which point it suffices to show that $g(f(\mathsf{refl}_{(w_1, w_2)})) = \mathsf{refl}_{(w_1, w_2)}$, which is true by definition.

Thus, f has a quasi-inverse, and is therefore an equivalence. □